JAX-sklearn (xlearn) — 五层架构 · 透明加速数据流

100% sklearn API → runtime injection → proxy dispatch → array conversion → JAX hardware

主数据流　　虚线框 = 架构分区

回退 / 反馈　　盒内 = 模块 + 职责

USER CODE · sklearn-compatible API

import xlearn as sklearn
LinearRegression · KMeans · PCA

Estimator Calls
fit · predict · transform

COMPATIBILITY · xlearn/__init__.py

__getattr__
lazy module loading

_auto_jax_accelerate_module
runtime class injection

AcceleratorRegistry
algorithm implementation map

JAX ACCELERATION · xlearn/_jax

EstimatorProxy
fit-time dispatch

Universal JAX
JIT · vmap · accelerated estimator

sklearn Fallback
original estimator

graceful fallback

DATA MANAGEMENT · boundary conversion

NumPy → JAX
dtype · shape normalization

Policy / Thresholds
dataset size · platform · config

JAX → NumPy
sklearn-compatible outputs

HARDWARE ABSTRACTION · JAX backend

CPU
jaxlib

CUDA GPU
JAX XLA

TPU
JAX XLA

Apple Metal
experimental

代理根据算法、数据规模和配置选择 JAX；不满足条件或执行失败时回退原生 sklearn。